Summary of the Driving-Factor Selection Process for p

1 Summary

This document selects variables for estimating the uncertain robotaxi choice share p in an ER-DRO model. The sequence is candidates → criteria → application of criteria → selection results.

$$\text{Observed value of } p_c = y_c = \frac{\text{Robotaxi trips in city } c}{M_c}.$$

The denominator M_c and the numerator must use the same geographic boundary, period, and trip unit. If usage data include only completed trips, y_c is a realized usage share that reflects supply conditions.

The final variable-selection table distinguishes exogenous variables from endogenous variables (dependent on operating decisions). When decision-dependent effects are incorporated into optimization, they must be connected to a decision-dependent model.

Future extensions to $p = f(x)+Q(x)\epsilon$ and decision dependence are also considered. This document incorporates only the conditions required for those extensions into the selection criteria.

Conclusion. Preliminary selection: "share of zero-vehicle households" and "age composition" as exogenous variables; "relative fare," "relative waiting time," and "in-vehicle support features" as endogenous variables.

2 Previously Established Candidates

2.1 Driving factors of p

The following table contains 23 candidate driving factors taken from the driving factors summarized in `A.Background Research`. This does not mean that all effects on city-level p have been established.

Variable	Supporting paper	Scope and results established in the paper	Measurement data
Relative fare	Yin and Cherchi (2024), *Willingness to pay for automated taxis*	Choice experiment between conventional and automated taxis. Analyzes choice differences by fare. Own-service fare is an operator decision.	Offered fares of both services + choice outcomes
Relative waiting time	Yin and Cherchi (2024)	Analyzes choice differences by waiting time. Own-service waiting time depends on fleet size and allocation.	Waiting times of both services + choice outcomes
In-vehicle travel time	Yin and Cherchi (2024)	Analyzes choice trade-offs between fare and travel time.	Offered travel times of both services
Trust, price value, and usage habits	Wei et al. (2024), *Using the Extended UTAUT*; Yin, Cherchi, and Nettle (2023)	Survey evidence on acceptance intentions and usage behavior. Wei does not include a trust variable. Attitude scores themselves are not choice probabilities.	Individual attitude and habit surveys
Age and robotaxi-related knowledge	Yin and Cherchi (2024); Wei et al. (2024)	A simple age effect is difficult to establish. Distinguish general AV knowledge from awareness of actual service operation.	Link individual characteristics to choice data
User reviews and in-vehicle support features	Yin and Cherchi (2024)	Analyzes choice effects of service attributes such as good reviews, destination changes, and communication with an operator.	Service attributes in the choice experiment
Robotaxi usage experience	Yin, Cherchi, and Nettle (2023), *Living in a City Where Automated Taxis are Operating*	Relationship of residence in an operating city and individual usage experience with preferences. A simple positive experience effect is not established.	Individual usage experience; city operation and launch records
Competitiveness of alternatives such as public transport	Cats et al., *Beyond the dichotomy*	Evidence from conventional ride-hailing. Substitution and complementarity coexist depending on travel-time competitiveness.	Alternative availability, fares, and times; GTFS, NTD
Share of zero-vehicle households	Elkhouly and Alhadidi; Loa, Lee, and Circella (2026)	Positive (+) association with conventional ride-hailing usage in Chicago districts. The attached document also reports a positive (+) association with robotaxi trip productions in SF.	ACS B08201

Variable	Supporting paper	Scope and results established in the paper	Measurement data
Income	Elkhouly and Alhadidi; Loa, Lee, and Circella (2026); Yin and Cherchi (2024); Wei et al. (2024)	Positive (+) in aggregate Chicago and SF data. The same effect is not consistently established in choice experiments and surveys.	ACS B19013
Employment and economic activity	Elkhouly and Alhadidi; Loa, Lee, and Circella (2026)	Economic activity share is positively (+) associated with conventional ride-hailing usage in Chicago; employment density with robotaxi trip attractions in SF.	LEHD/LODES; ACS B23025
Population density	Elkhouly and Alhadidi; Zhang et al.; Loa, Lee, and Circella (2026)	Spatial differences for conventional ride-hailing; a negative (-) association with robotaxi usage in SF.	ACS population + area
Land-use mix, accessibility, and commercial activity	Zhang et al.; Loa, Lee, and Circella (2026)	Nonlinear and spatially heterogeneous relationships in Chicago. A positive (+) association with land-use mix in SF.	EPA SLD
Public transport and walking environment	Zhang et al.; Cats et al.; Loa, Lee, and Circella (2026)	Substitution and complementarity coexist for conventional ride-hailing. The attached document reports positive (+) associations with frequency and walkability in SF.	EPA SLD; GTFS; NTD
Restaurant and leisure-facility density	Loa, Lee, and Circella (2026)	The attached document reports a positive (+) association with robotaxi trips in SF. The result does not separate the effects of M and p.	Census CBP; OpenStreetMap
Age composition	Loa, Lee, and Circella (2026); Etminani-Ghasrodashti et al. (2022)	Evidence on younger users in the attached document and conversation. Individual choice experiments do not readily establish a simple sign for age.	ACS B01001
Student share	Etminani-Ghasrodashti et al. (2022)	Usage evidence from the Arlington campus pilot. Free-ride support for students must be considered.	ACS enrollment data; local student population
Education and technology affinity	SAV choice-experiment and review literature from the previous conversation	Additional candidate. Educational attainment and technology affinity are different concepts; specific effects require confirmation.	ACS education data; technology-affinity surveys
Existing ride-hailing habits	Wei et al. (2024); previous conventional ride-hailing studies	Habits and past use may be connected to adoption. Past usage is a proxy for habits.	Common surveys; past conventional ride-hailing usage data
Safety perceptions and exposure	SAV studies from the previous conversation; Khan (2022) requires bibliographic and result verification	Candidate concerning perceived safety. Distinguish perceived safety from crash and crime exposure.	Safety surveys; crash and exposure data
Regulatory permission	Feasibility information rather than a paper on demand effects	Operation is not possible unless permission requirements are met. Treat separately from preference effects.	Permits and operating conditions from the relevant authorities; NCSL
Climate and ODD	Operating-feasibility conditions from the previous conversation	Compatibility between weather and the operating domain. Distinguish preference from service availability.	NOAA weather; operator ODD
Time since service launch	Yin, Cherchi, and Nettle (2023); launch and operation records	Candidate representing accumulated awareness and experience. The effect of elapsed time itself requires separate validation.	Company announcements; permit and launch records

3 Selection Criteria for the Driving Factors of p

No.	Criterion	Type
0	The dependent variable uses matching denominators, geographic boundaries, periods, and trip units	Before
1	Literature evidence relevant to p and an interpretable rationale exist	Before + after
2	Measurable at the decision time for all candidate cities using the same definition and source	Before
3	Interpretable; minimize redundant variables; limit variable count according to sample size	Before
4	Measure shares, densities, or relative service conditions rather than city size	Before
5	Distinguish dependence on operating decisions and define the role in the model	Before

No.	Criterion	Type
6	Candidate-city values match the range and service conditions of the training data	Before + after
7	No unexplained covariate-dependent patterns remain in the residuals after fitting	After
8	Data exist to estimate the response of p to decision changes in a decision-dependent extension	When extending

3.1 Discussion of Each Criterion

0. **Consistent dependent variable.** Match the numerator and denominator of robotaxi trips/M. Choice probability between two taxi types differs from p over all travel opportunities.

1. **Literature evidence.** Distinguish actual robotaxi, SAV pilot, and conventional ride-hailing evidence. If signs differ, examine service and regional differences and defer selection.

2. **Common measurement.** Use the same city boundaries, years, and variable definitions. Defer trust, safety perceptions, and similar variables without a common survey until data are available.

3. **Parsimony.** Start with one representative variable for similar concepts. Consider the number of independent cities; use $n/10$ only as a conservative reference rule of thumb.

4. **Variable form.** Population is the scale factor in M. For p, use zero-vehicle household share, facility density, relative fare, relative time, and similar measures.

5. **Exogenous/endogenous distinction.** Label city characteristics as exogenous candidates and service conditions set by the operator or affected by decisions as endogenous candidates. Do not exclude a candidate solely because it is endogenous.

6. **Transfer across cities.** Check whether district-level or other-country findings apply to candidate cities. Label values outside the training range as extrapolation.

7. **Residual checks.** For $f(x) + \epsilon$, examine residuals; for $f(x) + Q(x)\epsilon$, examine standardized residuals divided by $Q(x)$. Check distributional patterns as well as the mean and spread.

8. **Decision-dependent extension.** Changes in fare, fleet size, and other decisions must be accompanied by choice and service outcomes. If waiting time also responds to demand, account for that relationship jointly.

4 Applying the Criteria to the Candidates: p

The format matches the M application table in the attached document: **variable / 1 / 2 / 3 / 4 / 5 / decision / reason and form of use**. Criterion 0 is a common prerequisite; 6, 7, and 8 require data and fitted results and will be checked later.

○: applicable by design; △: further confirmation; –: not applicable. The + and - signs under criterion 1 indicate the associations reported in the literature. Under criterion 5, Ex., En., Mix., and Sep. mean exogenous, endogenous, depends on the component, and separate condition, respectively. **Selection is preliminary and does not mean data validation is complete.**

Variable	1	2	3	4	5	Decision	Reason / form of use
Relative fare	-	△	○	○	Mix.	Select	Own fare is endogenous; competing fare is an exogenous candidate. Use the fare difference for the same trip and incorporate it into the decision-dependent extension.
Relative waiting time	-	△	○	○	Mix.	Select	Own waiting time depends on fleet size and allocation. Compare with competing services and incorporate it into the decision-dependent extension.
In-vehicle travel time	-	△	○	○	Mix.	Defer	Traffic conditions are exogenous; the component affected by routes and operating policies is endogenous. Relative travel-time data require confirmation.
Trust, price value, and usage habits	△	△	△	○	Mix.	Defer	Common surveys are needed. Do not combine different concepts into one variable.

Variable	1	2	3	4	5	Decision	Reason / form of use
Age and related knowledge	△	△	△	○	Ex.	Partial	Merge age into age composition in C. Defer operating awareness and knowledge until survey data are checked.
Reviews and in-vehicle support features	+	△	△	○	Mix.	Partial	Select support features as endogenous candidates. Defer reviews separately because they are an outcome of past usage.
Robotaxi usage experience	△	△	○	○	Mix.	Defer	Check that experience predates the current decision. Do not directly replace individual experience with city service duration.
Competitiveness of alternatives	△	△	△	○	Ex.	Defer	Compare alternative fares, times, and availability. Check overlap with public transport and the walking environment.
Share of zero-vehicle households	+	○	○	○	Ex.	Select	Share of households without vehicles. Directly explains constraints on the private-car alternative and is commonly measurable through ACS.
Income	△	○	△	○	Ex.	Defer	Aggregate usage and choice experiments do not yield consistent effects. Add income or economic activity share one at a time and compare.
Employment and economic activity	+	○	△	○	Ex.	Defer	Specify either economic activity share or employment density. Check overlap with income.
Population density	△	○	△	○	Ex.	Defer	Regional differences in effects. Check overlap with zero-vehicle households and public transport.
Land-use mix and accessibility	△	△	△	○	Ex.	Defer	Select one representative indicator and check how district measures are aggregated to cities.
Public transport and walking environment	△	△	△	○	Ex.	Defer	Substitution and complementarity coexist. Check overlap with alternative competitiveness and zero-vehicle households.
Restaurant and leisure-facility density	+	△	○	○	Ex.	Defer	Usage evidence from one city, SF. Examine after checking facility definitions and transfer across cities.
Age composition	△	○	○	○	Ex.	Select	Initially consider the share aged 18–34. Evidence exists on younger users, but the city-level sign must be checked later.
Student share	△	△	△	○	Ex.	Defer	Overlaps with age. Campus and student fare-support effects are mixed in.
Education and technology affinity	△	△	△	○	Mix.	Defer	Distinguish educational attainment from technology affinity. Direct surveys and supporting evidence require confirmation.
Existing ride-hailing habits	△	△	△	○	Mix.	Defer	Conventional ride-hailing data before the decision are needed. Distinguish past usage from attitudinal habits.
Safety perceptions and exposure	△	△	△	○	Mix.	Defer	Common safety surveys are needed. Distinguish post-choice satisfaction from crash and crime exposure.
Regulatory permission	–	△	–	–	Sep.	Separate	Treat as an operating-feasibility condition rather than a choice-probability variable.
Climate and ODD	△	△	△	○	Mix.	Separate	Climate is exogenous. Assess ODD and service availability according to technical and operating conditions.
Time since service launch	△	△	○	○	En.	Defer	Launch timing depends on operating decisions. Consider in an extension addressing the post-entry time path.

5 Preliminary Selection Results: Exogenous/Endogenous Labels

Two city characteristics and three decision-dependent service variables are identified as initial priorities. Endogenous service variables will be incorporated into the decision-dependent ER-DRO extension after criterion 8 is checked. The following distinction is based on dependence on operating decisions.

Variable	Supporting paper	Scope and results established in the paper	Measurement data
Share of zero-vehicle households (exogenous)	Elkhouly and Alhadidi; Loa, Lee, and Circella (2026)	Represents constraints on the private-car alternative. Use zero-vehicle households / all households.	ACS B08201

Variable	Supporting paper	Scope and results established in the paper	Measurement data
Age composition (exogenous)	Loa, Lee, and Circella (2026); Etminani-Ghasrodashti et al. (2022)	Represents age composition. Use population aged 18–34 / total population. Do not uniformly fix the sign for individual choice.	ACS B01001
Relative fare (own fare: endogenous)	Yin and Cherchi (2024)	Choice response to fare differences. Selected for a decision-dependent extension that changes own fare.	Fares of both services + choice outcomes
Relative waiting time (own waiting time: endogenous)	Yin and Cherchi (2024)	Choice response to waiting-time differences. Connect fleet size and allocation to waiting time.	Fleet, allocation, and waiting-time records + choice outcomes
In-vehicle support features (endogenous candidate)	Yin and Cherchi (2024)	Choice effects of destination changes, operator contact, and similar features. Retained for an operating extension that selects features.	Feature-specific choice data

Further consideration. Income or economic activity share is the next additional candidate. Determine the number of variables based on observed city count and overlap. Treat competing-service fares and waiting times as exogenous candidates, but check the observation range. If fares and features are not decision variables in the current model, the endogenous labels indicate their roles in an extension.

Separate treatment. Treat regulation as operating feasibility and climate/ODD as service availability. Defer trust, knowledge, experience, safety, and similar variables until common data are available. Consider elapsed service time in a post-entry extension.

Post-selection checks. After gathering data, validate transfer across cities (6), residual patterns (7), and decision responses (8) to finalize selection. For $f(x) + Q(x)\epsilon$, do not automatically reject a variable because the original residuals are heteroscedastic; examine standardized residuals.

6 Remaining Tasks / Limitations

- Estimating p requires observed values (y), namely robotaxi trip-count data for city c.
- **Observed p and demand D:** p calculated from completed trips includes the effects of supply shortages. Using it directly as demand may underestimate unrealized demand. Adjustment using request/rejection data or an explicit assumption that supply constraints are small is needed.
- Further data checks are needed for the exogenous/endogenous variables of p.